



Mauricio Ríos Hernández

Mechanical Engineer • Mechanical Design • Mechanical Projects • Personnel coordinator • CAD & CAM • FEA
• Electronics & Intrumentation • Embedded Systems • Automation • Programming & DataScience • Physics

Contact

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GitHub: github.com/mrioshe
GrabCAD: grabcad.com/mauricioRios

Technical Skills

- CADs: • Inventor • AutoCAD • Fusion
- FEA: • ANSYS (Static Structural, Fluent)
- Circuits simulation: • LTspice
- OrCAD • Proteus • Digital
- Programming: • Python • MATLAB • LabVIEW
- Java • SQL • C & C++ •Linux
- R & RStudio • LaTeX • EES
- MCUs: • STM32 • Arduino • ESP32
- PLCs: • TIA Portal • ISPSoft & COMMGR
- Office: • Word • Excel • Power Point • Acces

Languages & Mobility

Spanish: Native
English: B1 (intermediate)
Driving licenses: A2 & B1 • Own vehicle • Full availability

Education

- B.Sc. Mechanical Engineering**
University of Antioquia • Feb 2017 – Sep 2024
- B.Sc. (Studying) — Physics Engineering**
National University of Colombia • Aug 2020 – Present
- B.Sc. (Studying) — Applied Data Science with Python Specialization**
Online Coursera Specialization • University Of Michigan
- Technical Degree — Software Programming**
Servicio Nacional de Aprendizaje (SENA) • Jan 2015 – Nov 2016

Professional Summary

Mechanical Engineer with a strong focus on mechanical design and electronic systems integration. Versatile and quick to adapt to new responsibilities, effective both individually and in teams. Experienced in taking engineering projects from quotation through commissioning and final delivery. Practical experience in automated pneumatic machinery design, CAD-based part and assembly modeling, finite element simulation, and embedded control.

Professional Experience

Resident Engineer & Mechanical Project Manager
Proymant Proyectos y Mantenimientos S.A.S. • Aug 2024 – Present

- Coordinated personnel for the execution of corrective and preventive maintenance tasks, as well as the implementation of metal-mechanical projects at Eurocerámica’s production plant.
- Led mechanical project development for industrial companies, covering the full cycle from quotation and commissioning to final delivery
- Provided drafting services for Eurocerámica, including machine component drawings, manufacturing plans, layout designs, mechanical concepts, and finite element simulations.
- Highlighted Mechanical Projects: Silo enclosure system and containment structure for truck scale (fosobáscula).

Mechanical Engineering Intern
Crystal S.A.S. • Feb 2024 – Aug 2024

- Supported mechanical design and assembly tasks for production equipment.
- Participated in prototyping and testing phases.
- Design a semi-automatic machine for folding, gluing and plastic-strapping of cardboard label-stands (design & prototyping).
- Design and construction of a labeling and plastic-strapping machine for sock packages using tagging machines.
- Design of a three-head marking machine design.

Own engineering projects

- Development of a compact CNC engraver using a low-power diode laser. The project covers the full design process—from current source and control circuitry to firmware implementation and wireless connectivity
- Development of a capacitive sensor to measure and control the fluid level in a tank.
- Design and Construction (Machining) of a Standing Wave Generator on a String for a Wave Physics Laboratory
- Development of PID Temperature Control Using ESP32 for the Heating Module of a Sublimation Press.
- AM signal modulator was developed using an ESP32 and Python, including both the circuit design and firmware. Additionally, a separate system was built using Arduino for signal demodulation, acquisition, and visualization.

Physics Laboratory Teaching Assistant — Electricity & Magnetism
National University of Colombia • Feb 2023 – Nov 2023

- Delivered theoretical instruction aligned with laboratory practices and supported students during experiments.
- Repaired and maintained laboratory equipment and contributed to experiment design.

